

PDF BOOK VERIFIER

Verification Report

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434

TOTAL

1

CORRECT

1

ERRORS

432

UNCERTAIN

Accuracy: 0.2% | 1 issue(s) found

Error Summary (1 issues)

1. [Page 13 | Line 370] $x = x + 5$

-> Equation appears incorrect. LHS - RHS evaluates to -5 (should be 0). Simplified difference: -5

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UNCERTAIN	general_text	Ln 1	Part 1
UNCERTAIN	general_text	Ln 2	Programming Environment, Variables,
UNCERTAIN	general_text	Ln 3	Expressions & Statements
UNCERTAIN	general_text	Ln 4	1.1
UNCERTAIN	general_text	Ln 5	Unit Overview & Learning Objectives
UNCERTAIN	general_text	Ln 6	Core Concept: Unit 1 Academic Scope
UNCERTAIN	general_text	Ln 7	[LPU CORE]
UNCERTAIN	general_text	Ln 8	This unit introduces the foundational architecture of the Python programming lan...
UNCERTAIN	general_text	Ln 9	a first-year engineering student, you will master the mechanics of the Python Vi...
UNCERTAIN	general_text	Ln 10	(PVM), understand the critical distinctions between Python 2 and Python 3, confi...
UNCERTAIN	general_text	Ln 11	variables, execute code in interactive and script modes, and manipulate data usi...
UNCERTAIN	general_text	Ln 12	typed variables, operators, expressions, formatted I/O, and string operations.
UNCERTAIN	general_text	Ln 13	? Environment Architecture: Configure Python on local systems, understand the PA...
UNCERTAIN	general_text	Ln 14	and distinguish bytecode compilation from virtual machine interpretation.
UNCERTAIN	general_text	Ln 15	? Variables & Dynamic Memory: Understand Python's reference model, identifier na...
UNCERTAIN	general_text	Ln 16	small integer caching, and keyword constraints.
UNCERTAIN	general_text	Ln 17	? Type System & Casting: Inspect and manipulate primitive types (int, float, str...
UNCERTAIN	general_text	Ln 18	the type() and explicit constructor functions.
UNCERTAIN	general_text	Ln 19	? Operators & Precedence: Apply arithmetic, relational, logical, assignment, and...
UNCERTAIN	general_text	Ln 20	with precision, mastering PEMDAS evaluation and right-to-left exponentiation.
UNCERTAIN	general_text	Ln 21	? Formatted Input/Output: Utilize print() formatting parameters (sep, end) and m...
UNCERTAIN	general_text	Ln 22	f-string interpolation with alignment and precision specifiers.
UNCERTAIN	general_text	Ln 23	? String Operations & Composition: Perform string concatenation and repetition, ...
UNCERTAIN	general_text	Ln 24	nested expressions, and document code effectively with comments and docstrings.
UNCERTAIN	general_text	Ln 25	1.2
UNCERTAIN	general_text	Ln 26	Setting up the Programming Environment
UNCERTAIN	general_text	Ln 27	1.2.1
UNCERTAIN	general_text	Ln 28	Python Architecture: Compilation vs. Interpretation
UNCERTAIN	factual_claim	Ln 29	Python is classified as a high-level, interpreted programming language, but inte...
UNCERTAIN	general_text	Ln 30	two-stage hybrid execution model:
UNCERTAIN	factual_claim	Ln 31	1. Source Code Compilation to Bytecode: When a Python source file (.py) is execu...
UNCERTAIN	general_text	Ln 32	CPython compiler translates human-readable source code into platform-independent...
UNCERTAIN	factual_claim	Ln 33	instructions known as bytecode (saved in .pyc files inside the __pycache__ folde...
UNCERTAIN	general_text	Ln 34	2. Bytecode Interpretation via PVM: The **Python Virtual Machine (PVM)** reads t...
UNCERTAIN	general_text	Ln 35	instructions line-by-line and translates them into machine-level instructions fo...

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UNCERTAIN	general_text	Ln 37	Chapter 1.
UNCERTAIN	general_text	Ln 38	Programming Environment, Variables, Expressions & Statements
UNCERTAIN	general_text	Ln 40	Python Source Code
UNCERTAIN	general_text	Ln 41	(program.py)
UNCERTAIN	general_text	Ln 42	CPython Bytecode
UNCERTAIN	general_text	Ln 43	(program.pyc)
UNCERTAIN	general_text	Ln 44	Python Virtual
UNCERTAIN	general_text	Ln 45	Machine (PVM)
UNCERTAIN	general_text	Ln 46	Host CPU
UNCERTAIN	general_text	Ln 47	Execution
UNCERTAIN	general_text	Ln 48	Compiler
UNCERTAIN	general_text	Ln 49	Interpreter
UNCERTAIN	general_text	Ln 50	Native Code
UNCERTAIN	general_text	Ln 51	Figure 1.1: The CPython Two-Stage Execution Lifecycle.
UNCERTAIN	general_text	Ln 52	1.2.2
UNCERTAIN	general_text	Ln 53	Python 2 vs. Python 3: Core Differences
UNCERTAIN	general_text	Ln 54	Feature / Behavior
UNCERTAIN	general_text	Ln 55	Python 2 (Legacy -- Deprecated
UNCERTAIN	general_text	Ln 56	2020)
UNCERTAIN	general_text	Ln 57	Python 3 (Modern University
UNCERTAIN	general_text	Ln 58	Standard)
UNCERTAIN	general_text	Ln 59	Print Syntax
UNCERTAIN	general_text	Ln 60	Statement without parentheses: print
UNCERTAIN	general_text	Ln 61	"Hello"
UNCERTAIN	general_text	Ln 62	Built-in function: print("Hello")
UNCERTAIN	general_text	Ln 63	Integer Division
UNCERTAIN	general_text	Ln 64	Truncating floor division: 5 / 2 yields 2
UNCERTAIN	general_text	Ln 65	True float division: 5 / 2 yields 2.5
UNCERTAIN	general_text	Ln 66	String Encoding
UNCERTAIN	general_text	Ln 67	ASCII by default; Unicode requires
UNCERTAIN	general_text	Ln 68	u"..."
UNCERTAIN	factual_claim	Ln 69	All strings are Unicode (UTF-8) by
UNCERTAIN	general_text	Ln 70	default
UNCERTAIN	general_text	Ln 71	User Input
UNCERTAIN	general_text	Ln 72	raw_input() for strings; input()
UNCERTAIN	general_text	Ln 73	evaluates
UNCERTAIN	general_text	Ln 74	Unified input() always returns a str
UNCERTAIN	general_text	Ln 75	Range Generator

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UNCERTAIN	general_text	Ln 76	range() creates an in-memory list;
UNCERTAIN	general_text	Ln 77	xrange()
UNCERTAIN	factual_claim	Ln 78	range() is an efficient generator
UNCERTAIN	general_text	Ln 79	sequence object
UNCERTAIN	general_text	Ln 80	Table 1.1: Key Technical Distinctions Between Python 2 and Python 3.
UNCERTAIN	general_text	Ln 81	1.2.3
UNCERTAIN	general_text	Ln 82	Windows Installation & The PATH Environment Variable
UNCERTAIN	factual_claim	Ln 83	When installing Python on Windows, checking ""Add Python to PATH"" is essentia...
UNCERTAIN	general_text	Ln 84	Core Concept: Why PATH Configuration Matters
UNCERTAIN	factual_claim	Ln 85	The PATH is an operating system environment variable that specifies directories ...
UNCERTAIN	general_text	Ln 86	executable binaries. Adding Python to PATH ensures that typing python or pip in ...
UNCERTAIN	general_text	Ln 87	Prompt (CMD) or PowerShell terminal allows the OS to immediately locate and laun...
UNCERTAIN	general_text	Ln 88	preter without specifying its full path (e.g., C:/Users/AppData/Local/Programs/P...
UNCERTAIN	general_text	Ln 89	To verify installation and environment configuration via CLI:
UNCERTAIN	general_text	Ln 90	Terminal Output:
UNCERTAIN	general_text	Ln 91	C:/Users/Student> python -version Python 3.11.4
UNCERTAIN	inequality	Ln 92	C:/Users/Student> python -c "print('Environment configured successfully!')" Envi...
-> Inequality or unparseable expression -- deferred to AI verifier			
UNCERTAIN	general_text	Ln 93	configured successfully!
UNCERTAIN	general_text	Ln 94	1.2.4
UNCERTAIN	general_text	Ln 95	Running Python: Interactive Mode vs. Script Mode
UNCERTAIN	general_text	Ln 96	1. Interactive Mode (REPL): Launching python in terminal opens the ""Read-Eval-P...
UNCERTAIN	factual_claim	Ln 97	Expressions typed at the prompt are evaluated and printed immediately:
UNCERTAIN	general_text	Ln 98	Terminal Output:
UNCERTAIN	inequality	Ln 99	>>> 5 + 7 12 >>> type("LPU") <class 'str'>
-> Inequality or unparseable expression -- deferred to AI verifier			
UNCERTAIN	general_text	Ln 100	2. Script Mode: Writing code into a text file with a .py extension (e.g., hello....
UNCERTAIN	general_text	Ln 101	via python hello.py. Expressions evaluated on a line (e.g., 5 + 7) execute silen...
UNCERTAIN	general_text	Ln 102	in print().

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UNCERTAIN	general_text	Ln 103	Chapter 1.
UNCERTAIN	general_text	Ln 104	Programming Environment, Variables, Expressions & Statements
UNCERTAIN	general_text	Ln 106	1.3
UNCERTAIN	general_text	Ln 107	Variables: Naming, Dynamic Typing, and Keywords
UNCERTAIN	general_text	Ln 108	1.3.1
UNCERTAIN	general_text	Ln 109	Python's Dynamic Memory Model
UNCERTAIN	factual_claim	Ln 110	In Python, a **variable** is not a fixed memory container of a fixed type. Inste...
UNCERTAIN	general_text	Ln 111	**named reference (pointer/tag)** bound to an object in heap memory.
UNCERTAIN	general_text	Ln 112	Formal Python Syntax Specification: Variable Assignment
CORRECT	equation	Ln 113	variable_name = value_expression
-> Numerically verified at test point (symbolic simplification inconclusive)			
UNCERTAIN	general_text	Ln 116	Integer Object
UNCERTAIN	general_text	Ln 117	value:
UNCERTAIN	general_text	Ln 119	type:
UNCERTAIN	general_text	Ln 120	int
UNCERTAIN	general_text	Ln 121	String Object
UNCERTAIN	general_text	Ln 122	value:
UNCERTAIN	general_text	Ln 123	"Hello"
UNCERTAIN	general_text	Ln 124	type:
UNCERTAIN	general_text	Ln 125	str
UNCERTAIN	general_text	Ln 126	References
UNCERTAIN	general_text	Ln 127	References
UNCERTAIN	general_text	Ln 128	Figure 1.2: Python's Name-Binding Pointer Architecture.
UNCERTAIN	general_text	Ln 129	Program Listing : Dynamic Rebinding and Identity Inspection
UNCERTAIN	general_text	Ln 130	1 # Dynamic Typing in action
UNCERTAIN	equation	Ln 131	2 x = 100
-> Could not parse expression for algebraic check: '2 x = 100'			
UNCERTAIN	general_text	Ln 132	# x references an integer object
UNCERTAIN	equation	Ln 133	3 print("Initial x:", x, " Type:", type(x), " ID:", id(x))
-> Inequality or unparseable expression -- deferred to AI verifier			
UNCERTAIN	factual_claim	Ln 135	5 x = "Lovely University" # x is rebound to a string object
UNCERTAIN	equation	Ln 136	6 print("Rebound x:", x, " Type:", type(x), " ID:", id(x))
-> Inequality or unparseable expression -- deferred to AI verifier			
UNCERTAIN	general_text	Ln 137	1.3.2
UNCERTAIN	factual_claim	Ln 138	Object Identity vs. Equality (is vs ==)
UNCERTAIN	equation	Ln 139	? Equality Operator (==): Compares whether two objects hold the **same value/con...
-> Could not parse expression for algebraic check: '? Equality Operator (==): Compares whether two objects hold the **same value/content**.'			
UNCERTAIN	factual_claim	Ln 140	? Identity Operator (is): Compares whether two variables point to the **exact sa...
UNCERTAIN	equation	Ln 141	address** (id(a) == id(b)).
-> Could not parse expression for algebraic check: 'address** (id(a) == id(b)).'			

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UNCERTAIN	general_text	Ln 142	? Small Integer Caching (Integer Pooling): CPython pre-allocates and caches inte...
UNCERTAIN	general_text	Ln 143	the range [-5, 256]. Any variables assigned integers within this range point to ...
UNCERTAIN	general_text	Ln 144	memory object.
UNCERTAIN	general_text	Ln 145	Program Listing : Demonstrating Integer Pooling and Identity
UNCERTAIN	equation	Ln 146	1 a = 256 -> Could not parse expression for algebraic check: '1 a = 256'
UNCERTAIN	equation	Ln 147	2 b = 256 -> Could not parse expression for algebraic check: '2 b = 256'
UNCERTAIN	inequality	Ln 148	3 print("256: a == b ->", a == b, " a is b ->", a is b) # True, True (Cached) -> Inequality or unparseable expression -- deferred to AI verifier
UNCERTAIN	equation	Ln 150	5 x = 1000 -> Could not parse expression for algebraic check: '5 x = 1000'
UNCERTAIN	equation	Ln 151	6 y = 1000 -> Could not parse expression for algebraic check: '6 y = 1000'
UNCERTAIN	inequality	Ln 152	7 print("1000: x == y ->", x == y, " x is y ->", x is y) # True, False (Distinc... -> Inequality or unparseable expression -- deferred to AI verifier
UNCERTAIN	general_text	Ln 153	objects)
UNCERTAIN	general_text	Ln 154	Terminal Output:
UNCERTAIN	general_text	Ln 155	256:
UNCERTAIN	inequality	Ln 156	a == b -> True a is b -> True 1000: -> Inequality or unparseable expression -- deferred to AI verifier
UNCERTAIN	inequality	Ln 157	x == y -> True x is y -> False -> Inequality or unparseable expression -- deferred to AI verifier

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UNCERTAIN	general_text	Ln 158	Chapter 1.
UNCERTAIN	general_text	Ln 159	Programming Environment, Variables, Expressions & Statements
UNCERTAIN	general_text	Ln 161	1.3.3
UNCERTAIN	general_text	Ln 162	Variable Naming Rules & Conventions
UNCERTAIN	general_text	Ln 163	Core Concept: Identifier Construction Rules
UNCERTAIN	factual_claim	Ln 164	An **identifier** is a name used to identify a variable, function, class, or mod...
UNCERTAIN	general_text	Ln 165	1. Starting Character: Must begin with a letter (A-Z, a-z) or an underscore (_).
UNCERTAIN	general_text	Ln 166	2. Subsequent Characters: Can contain letters, digits (0-9), and underscores.
UNCERTAIN	factual_claim	Ln 167	3. No Digits at Start: 2nd_value is invalid and raises SyntaxError.
UNCERTAIN	factual_claim	Ln 168	4. No Special Symbols: Characters like @, \$, %, -, . are illegal.
UNCERTAIN	factual_claim	Ln 169	5. Case-Sensitivity: total, Total, and TOTAL are three distinct identifiers.
UNCERTAIN	general_text	Ln 170	6. Conventions: snake_case for variables/functions, ALL_CAPS for constants, Pasc...
UNCERTAIN	general_text	Ln 171	classes.
UNCERTAIN	general_text	Ln 172	1.3.4
UNCERTAIN	general_text	Ln 173	Python Keywords (Reserved Words)
UNCERTAIN	definition	Ln 174	Keywords are reserved words that define the language's grammar. They cannot be u...
UNCERTAIN	general_text	Ln 175	names.
UNCERTAIN	general_text	Ln 176	False
UNCERTAIN	general_text	Ln 177	None
UNCERTAIN	general_text	Ln 178	True
UNCERTAIN	general_text	Ln 179	and
UNCERTAIN	general_text	Ln 181	assert
UNCERTAIN	general_text	Ln 182	async
UNCERTAIN	general_text	Ln 183	await
UNCERTAIN	general_text	Ln 184	break
UNCERTAIN	general_text	Ln 185	class
UNCERTAIN	general_text	Ln 186	continue
UNCERTAIN	general_text	Ln 187	def
UNCERTAIN	general_text	Ln 188	del
UNCERTAIN	general_text	Ln 189	elif
UNCERTAIN	general_text	Ln 190	else
UNCERTAIN	general_text	Ln 191	except
UNCERTAIN	general_text	Ln 192	finally
UNCERTAIN	general_text	Ln 193	for
UNCERTAIN	general_text	Ln 194	from
UNCERTAIN	general_text	Ln 195	global
UNCERTAIN	general_text	Ln 197	import
UNCERTAIN	general_text	Ln 200	lambda

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UNCERTAIN	general_text	Ln 201	nonlocal
UNCERTAIN	general_text	Ln 202	not
UNCERTAIN	general_text	Ln 204	pass
UNCERTAIN	general_text	Ln 205	raise
UNCERTAIN	general_text	Ln 206	return
UNCERTAIN	general_text	Ln 207	try
UNCERTAIN	general_text	Ln 208	while
UNCERTAIN	general_text	Ln 209	with
UNCERTAIN	general_text	Ln 210	yield
UNCERTAIN	general_text	Ln 211	Table 1.2: The Complete Set of Python Reserved Keywords.
UNCERTAIN	general_text	Ln 212	1.3.5
UNCERTAIN	general_text	Ln 213	Understanding and Preventing NameError
UNCERTAIN	general_text	Ln 214	A NameError occurs when code attempts to access a variable identifier that has n...
UNCERTAIN	general_text	Ln 215	object in the current scope.
UNCERTAIN	general_text	Ln 216	Common Student Pitfall & Bug Alert: Causes of NameError
UNCERTAIN	equation	Ln 217	1. Misspelling: Defining radius = 5 and typing print(raduis).
-> Could not parse expression for algebraic check: '1. Misspelling: Defining radius = 5 and typing print(raduis).'			
UNCERTAIN	definition	Ln 218	2. Usage Before Assignment: Referencing a variable above the line where it is de...
UNCERTAIN	equation	Ln 219	3. Forgetting Quotes on Strings: Writing city = Phagwara instead of city = "Phag...
-> Could not parse expression for algebraic check: '3. Forgetting Quotes on Strings: Writing city = Phagwara instead of city = "Phagwara).'			
UNCERTAIN	general_text	Ln 220	1.4
UNCERTAIN	general_text	Ln 221	Values, Data Types, and Type Inspection
UNCERTAIN	general_text	Ln 222	1.4.1
UNCERTAIN	general_text	Ln 223	Fundamental Primitive Data Types

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UNCERTAIN	general_text	Ln 224	Chapter 1.
UNCERTAIN	general_text	Ln 225	Programming Environment, Variables, Expressions & Statements
UNCERTAIN	general_text	Ln 227	Data Type
UNCERTAIN	general_text	Ln 228	Class Name
UNCERTAIN	general_text	Ln 229	Description / Characteristics
UNCERTAIN	general_text	Ln 230	Example Literal
UNCERTAIN	general_text	Ln 231	Integer
UNCERTAIN	general_text	Ln 232	int
UNCERTAIN	general_text	Ln 233	Whole numbers of arbitrary precision (no overflow
UNCERTAIN	general_text	Ln 234	limit)
UNCERTAIN	general_text	Ln 235	42, -1000, 0
UNCERTAIN	general_text	Ln 236	Float
UNCERTAIN	general_text	Ln 237	float
UNCERTAIN	general_text	Ln 238	Double-precision 64-bit IEEE 754 floating-point
UNCERTAIN	general_text	Ln 239	numbers
UNCERTAIN	general_text	Ln 240	3.14159, -0.005, 2.0
UNCERTAIN	general_text	Ln 241	String
UNCERTAIN	general_text	Ln 242	str
UNCERTAIN	general_text	Ln 243	Immutable sequence of Unicode characters
UNCERTAIN	general_text	Ln 244	"Python", 'LPU'
UNCERTAIN	general_text	Ln 245	Boolean
UNCERTAIN	general_text	Ln 246	bool
UNCERTAIN	general_text	Ln 247	Logical truth values (True or False)
UNCERTAIN	general_text	Ln 248	True, False
UNCERTAIN	general_text	Ln 249	Table 1.3: Core Primitive Data Types in Python 3.
UNCERTAIN	general_text	Ln 250	1.4.2
UNCERTAIN	general_text	Ln 251	Type Inspection with type()
UNCERTAIN	general_text	Ln 252	Program Listing : Inspecting Data Types
UNCERTAIN	equation	Ln 253	1 a = 15
-> Could not parse expression for algebraic check: '1 a = 15'			
UNCERTAIN	equation	Ln 254	2 b = 15.0
-> Could not parse expression for algebraic check: '2 b = 15.0'			
UNCERTAIN	general_text	Ln 255	3 c = "15"
UNCERTAIN	equation	Ln 256	4 d = True
-> Could not parse expression for algebraic check: '4 d = True'			
UNCERTAIN	general_text	Ln 258	6 print("a:", type(a))
UNCERTAIN	general_text	Ln 259	7 print("b:", type(b))
UNCERTAIN	general_text	Ln 260	8 print("c:", type(c))
UNCERTAIN	general_text	Ln 261	9 print("d:", type(d))

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UNCERTAIN	general_text	Ln 262	Terminal Output:
UNCERTAIN	general_text	Ln 264	<class 'int'> b:
UNCERTAIN	general_text	Ln 265	<class 'float'> c:
UNCERTAIN	general_text	Ln 266	<class 'str'> d:
UNCERTAIN	general_text	Ln 267	<class 'bool'>
UNCERTAIN	general_text	Ln 268	1.5
UNCERTAIN	general_text	Ln 269	Statements, Arithmetic Operators, and Precedence
UNCERTAIN	general_text	Ln 270	1.5.1
UNCERTAIN	general_text	Ln 271	Arithmetic Operators
UNCERTAIN	general_text	Ln 272	Operator
UNCERTAIN	general_text	Ln 273	Name
UNCERTAIN	general_text	Ln 274	Mathematical Meaning
UNCERTAIN	general_text	Ln 275	Behavioral Details
UNCERTAIN	general_text	Ln 276	Example
UNCERTAIN	general_text	Ln 278	Addition
UNCERTAIN	general_text	Ln 279	$a + b$
UNCERTAIN	general_text	Ln 280	Sum of numeric values
UNCERTAIN	general_text	Ln 281	$7 + 3 \text{ ? } 10$
UNCERTAIN	general_text	Ln 283	Subtraction
UNCERTAIN	general_text	Ln 284	$a - b$
UNCERTAIN	general_text	Ln 285	Difference of numeric values
UNCERTAIN	general_text	Ln 286	$7 - 3 \text{ ? } 4$
UNCERTAIN	general_text	Ln 288	Multiplication
UNCERTAIN	general_text	Ln 289	$a \times b$
UNCERTAIN	general_text	Ln 290	Product of values
UNCERTAIN	general_text	Ln 291	$7 * 3 \text{ ? } 21$
UNCERTAIN	general_text	Ln 293	True Division
UNCERTAIN	general_text	Ln 294	a / b
UNCERTAIN	general_text	Ln 295	Always returns a float
UNCERTAIN	general_text	Ln 296	$7 / 2 \text{ ? } 3.5$
UNCERTAIN	general_text	Ln 298	Floor Division
UNCERTAIN	general_text	Ln 299	$?a/b?$
UNCERTAIN	general_text	Ln 300	Rounds quotient down to
UNCERTAIN	general_text	Ln 301	nearest integer
UNCERTAIN	general_text	Ln 302	$7 // 2 \text{ ? } 3$
UNCERTAIN	general_text	Ln 304	Modulus
UNCERTAIN	general_text	Ln 305	$a \text{ (mod } b)$
UNCERTAIN	general_text	Ln 306	Returns remainder of division
UNCERTAIN	general_text	Ln 307	$7 \% 3 \text{ ? } 1$

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UNCERTAIN	general_text	Ln 309	Exponentiation
UNCERTAIN	general_text	Ln 311	Raises base a to power b
UNCERTAIN	general_text	Ln 312	2 ** 3 ?8
UNCERTAIN	general_text	Ln 313	Table 1.4: Python Arithmetic Operators and Evaluation Mechanics.

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UNCERTAIN	general_text	Ln 314	Chapter 1.
UNCERTAIN	general_text	Ln 315	Programming Environment, Variables, Expressions & Statements
UNCERTAIN	general_text	Ln 317	Exam Memory Hook & University Trap Alert: Negative Floor Division & Modulus
UNCERTAIN	general_text	Ln 318	Python's floor division rounds toward negative infinity:
UNCERTAIN	equation	Ln 319	$? 7 // 2 = 3$ (since $3.5 ? 3$)
-> Could not parse expression for algebraic check: ' $? 7 // 2 = 3$ (since $3.5 ? 3$)'			
UNCERTAIN	equation	Ln 320	$? -7 // 2 = -4$ (since -3.5 rounded down is -4)
-> Could not parse expression for algebraic check: ' $? -7 // 2 = -4$ (since -3.5 rounded down is -4)'			
UNCERTAIN	equation	Ln 321	$? -7 \% 2 = 1$ (satisfying $a = (a//b) \times b + (a\%b) = ? -7 = (-4 \times 2) + 1$)
-> Could not parse expression for algebraic check: ' $? -7 \% 2 = 1$ (satisfying $a = (a//b) \times b + (a\%b) = ? -7 = (-4 \times 2) + 1$)'			
UNCERTAIN	general_text	Ln 322	1.5.2
UNCERTAIN	general_text	Ln 323	Bitwise and Assignment Operators
UNCERTAIN	general_text	Ln 324	Operator
UNCERTAIN	general_text	Ln 325	Category
UNCERTAIN	general_text	Ln 326	Description
UNCERTAIN	general_text	Ln 327	Example
UNCERTAIN	general_text	Ln 329	Bitwise AND
UNCERTAIN	general_text	Ln 330	Bit-by-bit AND operation
UNCERTAIN	general_text	Ln 331	$5 \& 3$
UNCERTAIN	general_text	Ln 332	$(01012 \& 00112 ?$
UNCERTAIN	equation	Ln 333	$00012 = 1)$
-> Could not parse expression for algebraic check: ' $00012 = 1$)'			
UNCERTAIN	general_text	Ln 335	Bitwise OR
UNCERTAIN	general_text	Ln 336	Bit-by-bit OR operation
UNCERTAIN	general_text	Ln 337	$5 3$
UNCERTAIN	general_text	Ln 338	$(01012 00112 ?$
UNCERTAIN	equation	Ln 339	$01112 = 7)$
-> Could not parse expression for algebraic check: ' $01112 = 7)$ '			
UNCERTAIN	general_text	Ln 341	Bitwise XOR
UNCERTAIN	general_text	Ln 342	Bit-by-bit XOR (1 if bits differ)
UNCERTAIN	general_text	Ln 343	$5 \wedge 3$
UNCERTAIN	general_text	Ln 344	$(01012$
UNCERTAIN	general_text	Ln 346	00112
UNCERTAIN	equation	Ln 348	$01102 = 6)$
-> Could not parse expression for algebraic check: ' $01102 = 6)$ '			
UNCERTAIN	general_text	Ln 350	Bitwise NOT
UNCERTAIN	equation	Ln 351	Inverts bits ($\sim x = -(x + 1)$)
-> Could not parse expression for algebraic check: ' $\text{Inverts bits } (\sim x = -(x + 1))$ '			
UNCERTAIN	general_text	Ln 352	$\sim 5 ? -6$
UNCERTAIN	general_text	Ln 354	Left Shift
UNCERTAIN	general_text	Ln 355	Shifts bits left ($x \times 2^n$)

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UNCERTAIN	general_text	Ln 356	3 « 2
UNCERTAIN	general_text	Ln 357	(3 x
UNCERTAIN	general_text	Ln 358	4 ?12)
UNCERTAIN	general_text	Ln 360	Right Shift
UNCERTAIN	general_text	Ln 361	Shifts bits right (x//2n)
UNCERTAIN	general_text	Ln 362	12 » 2
UNCERTAIN	general_text	Ln 363	(12//4 ?3)
UNCERTAIN	equation	Ln 364	+=, -=, *=

-> Could not parse expression for algebraic check: '+ =, -=, *='

UNCERTAIN	general_text	Ln 365	Augmented Assignment
UNCERTAIN	general_text	Ln 366	Computes operation and updates variable in place
UNCERTAIN	general_text	Ln 367	x += 5
UNCERTAIN	general_text	Ln 369	shorthand for
ERROR	equation	Ln 370	x = x + 5

-> Equation appears incorrect. LHS - RHS evaluates to -5 (should be 0). Simplified difference: -5

UNCERTAIN	general_text	Ln 371	Table 1.5: Bitwise and Augmented Assignment Operators.
UNCERTAIN	general_text	Ln 372	1.5.3
UNCERTAIN	general_text	Ln 373	Order of Operations (PEMDAS Rules)
UNCERTAIN	general_text	Ln 374	When multiple operators appear in an expression, Python resolves their order via...
UNCERTAIN	general_text	Ln 375	1. Parentheses (): Highest precedence.
UNCERTAIN	general_text	Ln 376	2. Exponentiation **: Evaluates **Right to Left**!
UNCERTAIN	equation	Ln 377	2 ** 3 ** 2 = ?2(32) = 29 = 512

-> Could not parse expression for algebraic check: '2 ** 3 ** 2 = ?2(32) = 29 = 512'

UNCERTAIN	general_text	Ln 378	3. Multiplication *, Division /, Floor Division //, Modulus %: Left to Right.
UNCERTAIN	general_text	Ln 379	4. Addition +, Subtraction -: Left to Right.
UNCERTAIN	general_text	Ln 380	5. Bitwise Shifts «, », Bitwise AND &, XOR ^, OR .
UNCERTAIN	inequality	Ln 381	6. Comparisons and Membership: ==, !=, <, >, in, is.

-> Inequality or unparseable expression -- deferred to AI verifier

UNCERTAIN	inequality	Ln 382	7. Logical Operators: not > and > or.
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-> Inequality or unparseable expression -- deferred to AI verifier

UNCERTAIN	general_text	Ln 383	1.6
UNCERTAIN	general_text	Ln 384	Formatted Input and Output Operations
UNCERTAIN	general_text	Ln 385	1.6.1
UNCERTAIN	general_text	Ln 386	The print() Function Parameters
UNCERTAIN	general_text	Ln 387	The built-in print() function accepts optional keyword parameters:

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UNCERTAIN	general_text	Ln 388	Chapter 1.
UNCERTAIN	general_text	Ln 389	Programming Environment, Variables, Expressions & Statements
UNCERTAIN	equation	Ln 391	? sep=string: String inserted between values (default is a single space " "). -> Could not parse expression for algebraic check: '? sep = string: String inserted between values (default is a single space " ").'
UNCERTAIN	equation	Ln 392	? end=string: String appended after the last value (default is a newline "\n"). -> Could not parse expression for algebraic check: '? end = string: String appended after the last value (default is a newline "\n").'
UNCERTAIN	general_text	Ln 393	Program Listing : Customizing print() with sep and end
UNCERTAIN	equation	Ln 394	1 print("INT108", "MTH165", "CSE111", sep=" ") -> Could not parse expression for algebraic check: '1 print("INT108", "MTH165", "CSE111", sep = " ")'
UNCERTAIN	equation	Ln 395	2 print("Score", end=": ") -> Could not parse expression for algebraic check: '2 print("Score", end = ": ")'
UNCERTAIN	general_text	Ln 396	3 print(95)
UNCERTAIN	general_text	Ln 397	Terminal Output:
UNCERTAIN	general_text	Ln 398	INT108 MTH165 CSE111 Score:
UNCERTAIN	general_text	Ln 400	1.6.2
UNCERTAIN	general_text	Ln 401	Modern Formatted String Literals (f-strings)
UNCERTAIN	general_text	Ln 402	Formatted string literals (f-strings) provide powerful formatting capabilities:
UNCERTAIN	general_text	Ln 403	Program Listing : Advanced f-string Formatting
UNCERTAIN	general_text	Ln 404	1 item = "Laptop"
UNCERTAIN	equation	Ln 405	2 price = 45999.5 -> Could not parse expression for algebraic check: '2 price = 45999.5'
UNCERTAIN	equation	Ln 406	3 discount = 0.15 -> Could not parse expression for algebraic check: '3 discount = 0.15'
UNCERTAIN	equation	Ln 407	4 final_price = price * (1 - discount) -> Could not parse expression for algebraic check: '4 final_price = price * (1 - discount)'
UNCERTAIN	general_text	Ln 409	6 # Float precision, alignment, and thousands separators
UNCERTAIN	general_text	Ln 410	7 print(f"Product:
UNCERTAIN	general_text	Ln 411	{item:>15}")
UNCERTAIN	general_text	Ln 412	8 print(f"Original: Rs. {price:12,.2f}")
UNCERTAIN	general_text	Ln 413	9 print(f"Discount: {discount * 100:11.1f}%")
UNCERTAIN	general_text	Ln 414	10 print(f"Final:
UNCERTAIN	general_text	Ln 415	Rs. {final_price:12,.2f}")
UNCERTAIN	general_text	Ln 416	Terminal Output:
UNCERTAIN	general_text	Ln 417	Product:
UNCERTAIN	general_text	Ln 418	Laptop Original:
UNCERTAIN	general_text	Ln 419	Rs.
UNCERTAIN	general_text	Ln 420	45,999.50 Discount:
UNCERTAIN	general_text	Ln 421	15.0Final:
UNCERTAIN	general_text	Ln 422	Rs.
UNCERTAIN	general_text	Ln 423	39,099.58
UNCERTAIN	general_text	Ln 424	1.7

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UNCERTAIN	general_text	Ln 425	String Operations, Composition, and Comments
UNCERTAIN	general_text	Ln 426	1.7.1
UNCERTAIN	general_text	Ln 427	String Concatenation (+) and Repetition (*)
UNCERTAIN	general_text	Ln 428	Program Listing : String Concatenation and Banner Repetition
UNCERTAIN	general_text	Ln 429	1 first_name = "Alan"
UNCERTAIN	general_text	Ln 430	2 last_name = "Turing"
UNCERTAIN	equation	Ln 431	3 full_name = first_name + " " + last_name
-> Could not parse expression for algebraic check: '3 full_name = first_name + " " + last_name'			
UNCERTAIN	general_text	Ln 432	4 print("Full Name:", full_name)
UNCERTAIN	equation	Ln 434	6 banner = "=" * 35
-> Could not parse expression for algebraic check: '6 banner = "=" * 35'			
UNCERTAIN	general_text	Ln 435	7 print(banner)
UNCERTAIN	general_text	Ln 436	8 print(" LPU COMPUTER SCIENCE & IT ")
UNCERTAIN	general_text	Ln 437	9 print(banner)
UNCERTAIN	general_text	Ln 438	1.7.2
UNCERTAIN	general_text	Ln 439	Composition & Comments
UNCERTAIN	general_text	Ln 440	? Composition: Nesting expressions directly inside function calls:

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UNCERTAIN	general_text	Ln 441	Chapter 1.
UNCERTAIN	general_text	Ln 442	Programming Environment, Variables, Expressions & Statements
UNCERTAIN	general_text	Ln 444	Program Listing : Composition Demonstration
UNCERTAIN	equation	Ln 445	1 print("Length of formatted string:", len(str(2 ** 100)))
-> Inequality or unparseable expression -- deferred to AI verifier			
UNCERTAIN	equation	Ln 446	? Comments: Single line (#), multi-line comments, and docstrings ("""...""").
-> Inequality or unparseable expression -- deferred to AI verifier			
UNCERTAIN	general_text	Ln 447	1.8
UNCERTAIN	general_text	Ln 448	Exam Tips & Common Student Pitfalls
UNCERTAIN	general_text	Ln 449	Exam Memory Hook & University Trap Alert: High-Yield Unit 1 Exam Insights
UNCERTAIN	factual_claim	Ln 450	? Division Output: In Python 3, / ALWAYS produces a float (4 / 2 is 2.0, not 2).
UNCERTAIN	general_text	Ln 451	? Exponentiation Associativity: x ** y ** z evaluates from right to left.
UNCERTAIN	general_text	Ln 452	? String Multiplying by Zero/Negative: Multiplying a string by 0 or a negative i...
UNCERTAIN	general_text	Ln 453	results in an empty string ("").
UNCERTAIN	general_text	Ln 454	? Input Return Type: The input() function ALWAYS returns a str. Arithmetic opera...
UNCERTAIN	general_text	Ln 455	on raw inputs fail unless converted via int() or float().
UNCERTAIN	general_text	Ln 456	1.9
UNCERTAIN	general_text	Ln 457	Output Prediction & Debugging Challenges
UNCERTAIN	general_text	Ln 458	Output Prediction Challenge: Tricky Syntax and Expression Evaluation
UNCERTAIN	general_text	Ln 459	Question: Predict the exact output of the following code snippet:
UNCERTAIN	equation	Ln 460	1 a = 16
-> Could not parse expression for algebraic check: '1 a = 16'			
UNCERTAIN	equation	Ln 461	2 b = 3
-> Could not parse expression for algebraic check: '2 b = 3'			
UNCERTAIN	general_text	Ln 462	3 print(a // b, a % b, a / b)
UNCERTAIN	general_text	Ln 463	4 x = "3"
UNCERTAIN	equation	Ln 464	5 y = 2
-> Could not parse expression for algebraic check: '5 y = 2'			
UNCERTAIN	general_text	Ln 465	6 print(x * y * 2)
UNCERTAIN	general_text	Ln 466	7 print(2 ** 2 ** 3)
UNCERTAIN	general_text	Ln 467	8 print(5 & 3, 5 3, 5 ^ 3)
UNCERTAIN	general_text	Ln 468	Answer & Tracing:
UNCERTAIN	equation	Ln 469	1. a // b is 16 // 3 = 5.
-> Could not parse expression for algebraic check: '1. a // b is 16 // 3 = 5.'			
UNCERTAIN	equation	Ln 470	2. a % b is 16 % 3 = 1.
-> Could not parse expression for algebraic check: '2. a % b is 16 % 3 = 1.'			
UNCERTAIN	equation	Ln 471	3. a / b is 16 / 3 = 5.333333333333333.
-> Could not parse expression for algebraic check: '3. a / b is 16 / 3 = 5.333333333333333.'			
UNCERTAIN	general_text	Ln 472	4. x * y gives "3" * 2 = "33". Then "33" * 2 gives "3333".
UNCERTAIN	equation	Ln 473	5. 2 ** 2 ** 3 evaluates right-to-left: 23 = 8, then 28 = 256.
-> Could not parse expression for algebraic check: '5. 2 ** 2 ** 3 evaluates right-to-left: 23 = 8, then 28 = 256.'			

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UNCERTAIN

equation

Ln 474

6. $5 \& 3 = 1$; $5|3 = 7$; $5 \ ? 3 = 6$.

-> Could not parse expression for algebraic check: '6. $5 \& 3 = 1$; $5|3 = 7$; $5 \ ? 3 = 6$.'